

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457814.7329 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.053
Transit depth (Rp/Rs)^2	2.06 +/- 1.53 [%]
Orbital Inclination (inc)	80.16 +/- 2.88
Airmass coefficient 1 (a1)	0.957 +/- 0.072
Airmass coefficient 2 (a2)	0.03 +/- 0.03
Scatter in the residuals of the lightcurve fit is	7.51 %
Best Comparison Star	#3 - [390, 136]
Optimal Method	PSF photometry
Transit Duration (day)	0.038 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.144 ± 0.053 vs 0.1594 (deviation 0.29σ)
Duration fit vs published	0.038 d vs 0.0512 d (deviation 0.78σ)
Mid-time O-C	3.0 min
Depth SNR	2.7
Residual scatter	7.51 %

Light curve

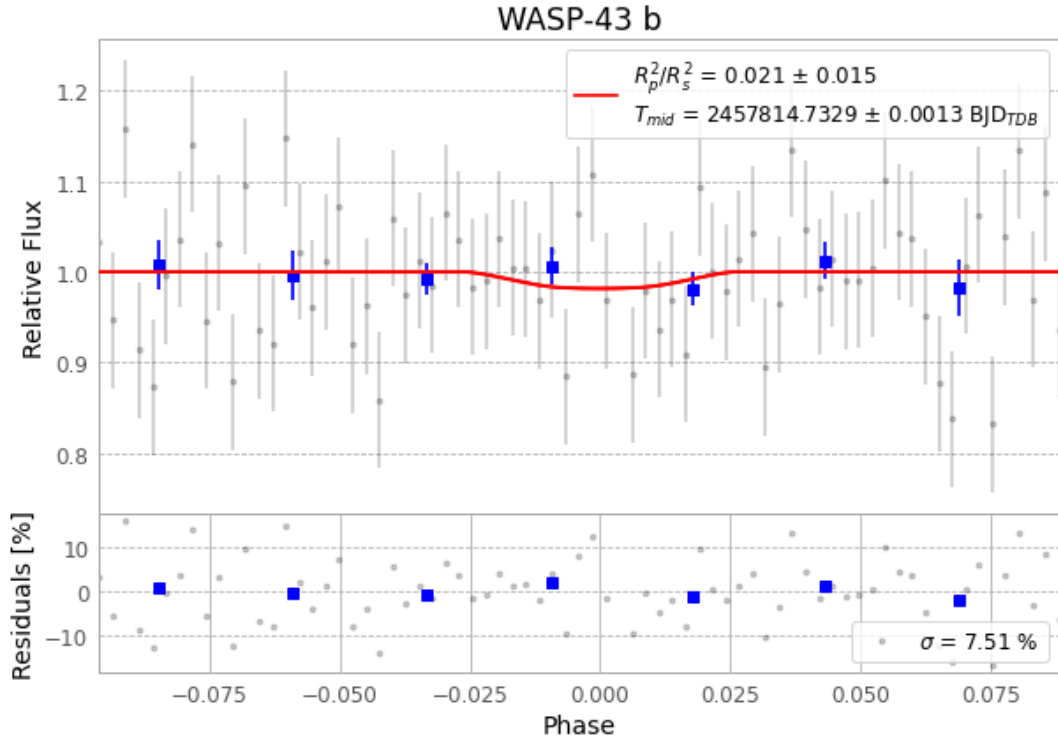


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-01.png (SHA-256 9b9323fdb9c532e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 147], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b33961f36c2b7c95...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

72 FITS frames (47 MB), WASP-43170302033312.FITS → WASP-43170302070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170302.log (273a4d557f70...), FinalLightCurve_WASP-43 b_2017-03-01.png (9b9323fdb9c...), FinalLightCurve_WASP-43 b_2017-03-01.csv (77e0bc46799c...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 22:31Z.